

Speech and Audio Systems

Project: Verified Study Buddy

1. Speech and Audio Systems

Teach the learning cycle

Primary target: Explain the mechanism and terminology of speech pipelines and multimodal evidence

Session · Sequence and minutes · Resources

Session: Session 1 · Understand · 45 min

Sequence and minutes: 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.

Resources: Sections 1–6; record the brief recall in the workbook.

Session: Session 2 · Practise · 45 min

Sequence and minutes: 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

Resources: Sections 7–10; use the matching workbook W2 and misconception check.

Session: Session 3 · Investigate and transfer · 45 min

Sequence and minutes: 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.

Resources: Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building. Repeated copies of a case share one recorded attempt; they are not new evidence.

Case exposure check: this source example appears in ai-2/26, ai-3/12, ai-3/13. Reuse a familiar case for

recall or modelling; verify that the case used to assess transfer has not already received feedback.

Retrieval answer guidance

Visual Composition

Without opening the old lesson, explain one key idea from Visual Composition. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the mechanism and terminology of image generation and provenance

Reference example, not the only acceptable wording: Teacher explanation: An image generator constructs an output from learned patterns and a specification. A generated picture is not a record that the depicted event occurred. Keep a provenance record containing specification, tool/version when known, generation date, edits, permissions and intended use. Label fictional imagery. Inspect visual requirements and avoid inferring identity or event authenticity from appearance.

Prompt Evaluation

Without opening the old lesson, explain one key idea from Prompt Evaluation. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the mechanism and terminology of controlled prompt experiments

Reference example, not the only acceptable wording: Teacher explanation: A prompt experiment compares complete specifications using the same task cases. Define success before inspecting outputs; change one intended instruction and preserve other settings. Evaluate both prompts on paired cases with fixed model/version, source records, output contract and scoring rubric. Use development results for selection, then reserve new final cases. Score each case rather than counting attractive sentences.

Examples and Rubrics

Without opening the old lesson, explain one key idea from Examples and Rubrics. Give an example and a mistake that the idea helps you avoid.

Look for: Write a prompt specification with observable response criteria

Reference example, not the only acceptable wording: Teacher explanation: Fields status,room,capacity,source_ids. Status supported or needs_review. supported requires matching current approval and current capacity for the same room; no unresolved conflict.

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

Audio says 'sixteen divided by four'; transcript says 'sixty divided by four'; calculator returns 15. Which stage first fails? What would a useful confirmation show?

Concept to explain

Speech interaction includes capture, recognition, interpretation, and response stages.

Term · Meaning

Term: capture

Meaning: Record a signal, observation or item of information for later use.

Term: speech recognition

Meaning: Converting spoken audio into an estimated sequence of words.

Term: intent

Meaning: The goal a user is trying to achieve, which may need clarification.

Term: synthesis

Meaning: Combining information into a new account or output while preserving its meaning and support.

Prepare the class

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Two 45-minute sessions. A multimodal pipeline may transcribe speech, inspect an image and generate a response. Errors at one stage can propagate even when later stages run correctly. Use the supplied constructed scenarios to make the mechanism inspectable. Collect W4 before sharing solutions. Students should name intermediate evidence and limitations; classroom pacing and learning effectiveness remain unvalidated.

Readiness answer

A result depends on the inputs, procedure and conditions that produced it. Preserve data roles and distinguish observed outcomes from predictions or unmeasured conditions. If the prior concept cannot be explained, use a short retrieval activity before this chapter.

Lesson and workbook; paper and pencil

Calculator and optional local Python 3 for arithmetic; no live API or private data

Prepare the fresh retry separately and withhold its solution until after collection

End goal for the pictorial case

By the end, I can separate speech recognition, calculation and speech output when diagnosing an error.

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

2. Explain the mechanism

Explain the mechanism

Input → Process → Output

Trace the information. At each stage, ask what changed and how you could test it.

Which stage could fail even when the other two are correct? Explain:

Predict first. Then check each step. A correct answer without visible reasoning is incomplete evidence.

1. Learner says 'define erosion' in a noisy room.
2. Recognizer outputs 'design erosion.'
3. Intent stage follows the wrong transcript.
4. Display transcript, allow correction, and avoid high-consequence action from uncertain audio.

Recalculate, retrace, or restate the reasoning in your own representation:

CHECK Did the evidence answer the exact question? Did the total, path, or source record stay consistent?

Concrete teaching case

A speech tool hears fifteen as fifty, then calculates correctly using fifty.

Where did the error enter?

Exact explanation to model

Transcription supplied the wrong number. Confirm the transcript or ask for clarification. Correct arithmetic cannot repair wrong input.

Explain the added worked example

A multimodal pipeline may transcribe speech, inspect an image and generate a response. Errors at one stage can propagate even when later stages run correctly.

Begin by writing the question the procedure is intended to answer. Identify the information already

supplied, the information it will produce and the information still missing. These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

Track source modality, segment/time references, transcription, interpretation and final claims separately. Compare conflicting evidence without assuming one modality always wins. Ask for clarification when the task needs unresolved information.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same inputs. If a required input is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Key terms

Term · Meaning

Term: Modality

Meaning: A form of information such as audio, image or text

Term: Transcription

Meaning: A text representation of speech

Term: Error propagation

Meaning: An earlier error affecting later processing

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a prediction, then explain the deciding step. These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Audio transcript reads fifteen boxes; a clear written inventory says fifty boxes. The original audio is unavailable and the assistant reports fifteen as certain.

There is an unresolved quantity conflict and no audio replay. Preserve both sources and ask the responsible person; certainty is unsupported.

Compare the intermediate evidence with the final statement. Explain why each number or decision follows from the stated inputs. The conclusion belongs to this scenario and procedure; changing the inputs, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Thinking move · Teacher prompt

Thinking move: Plan

Teacher prompt: What is given? What is the question? Which procedure fits these facts?

Thinking move: Do

Teacher prompt: Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.

Thinking move: Check

Teacher prompt: Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Speech and Audio Systems

A fictional learner dictates a calculation to a voice assistant.

The facts we will use

Known audio phrase: “thirteen plus seven”. Transcript: “thirty plus seven”. Calculator returns 37.

Synthetic voice says “thirty-seven”. The example supplies ground truth; no real audio was analysed.

Speech and Audio Systems

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Recognise speech

Transcription changes thirteen into thirty.



2. Calculate text

$30+7=37$ is correct for the mistaken transcript.



3. Speak result

Speech synthesis repeats 37; fluent voice does not validate the input.



4. Confirm intended input

Read back the recognised expression; intended $13+7=20$.

CHANGE ONE THING

Correct the transcript to "thirteen plus seven" with all later stages unchanged.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Recognise speech

What happens and why: Transcription changes thirteen into thirty.

Step: 2. Calculate text

What happens and why: $30+7=37$ is correct for the mistaken transcript.

Step: 3. Speak result

What happens and why: Speech synthesis repeats 37; fluent voice does not validate the input.

Step: 4. Confirm intended input

What happens and why: Read back the recognised expression; intended $13+7=20$.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

The first error occurs before arithmetic. Blaming the calculator would target a stage that correctly processed its supplied text. Read-back or on-screen confirmation lets the user catch a number misunderstanding before action. Retain the original audio and transcript, where permitted, to separate recognition defects from later processing errors.

What this example does not establish

These supplied transcript facts do not establish how an actual recogniser performs for accents, noise or different speakers.

Change one thing in the pictured case

Change: Correct the transcript to “thirteen plus seven” with all later stages unchanged.

Prediction: The calculator and spoken result should become 20 under the declared pipeline.

Reason: $13+7=20$; the changed input removes the earlier recognition error.

Ask while showing the picture

Cover the result. Ask learners to name the inputs and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its evidence. Then change the supplied condition and compare. The picture is a representation; it does not document execution of a real AI model.

3. Demonstrate and guide practice

Demonstrate and guide practice

Board sequence

Write the given facts: A speech tool hears fifteen as fifty, then calculates correctly using fifty.

Write the question to explain: Where did the error enter?

Reveal after predictions: Transcription supplied the wrong number. Confirm the transcript or ask for

clarification. Correct arithmetic cannot repair wrong input.

Keep for an independent attempt: A recording says fifteen, but the transcript says fifty. The summary repeats fifty. Where did the error enter, and what should be checked first?

Classroom dialogue

Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

In pairs, one learner solves while the other checks evidence; exchange roles. Remove the worked answer before independent assessment.

Model these guided questions

Guided: There is an unresolved quantity conflict and no audio replay. Preserve both sources and ask the responsible person; certainty is unsupported. Qualification: A stage-level accuracy or authority policy does not establish every modality agrees or every downstream claim is supported. Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the mechanism; reject a claim resting on unmeasured evidence.

Case-specific teaching sequence

Stage · Teacher action

Stage: Session 1: 0–7 minutes

Teacher action: Retrieve the prerequisite; identify the input, procedure and evidence source.

Stage: 7–18 minutes

Teacher action: Explain section 1 and vocabulary. Complete W1 with a mechanism-based explanation.

Stage: 18–32 minutes

Teacher action: Trace the declared procedure and worked case. Complete W2, recording intermediate steps.

Stage: 32–45 minutes

Teacher action: Discuss the diagram with the worked input; use W3 to distinguish measured claims and limitations.

Stage: Session 2: 0–8 minutes

Teacher action: Retrieve the worked reasoning and collect the guided response before discussing it.

Stage: 8–20 minutes

Teacher action: Compare a proposed boundary variant with the original case; identify what remains fixed.

Stage: 20–35 minutes

Teacher action: Collect W4 independently. Require the procedure trace and the bounded conclusion.

Stage: 35–45 minutes

Teacher action: Use W5 and exit responses for feedback. Administer the fresh retry after feedback and preserve both attempts.

Reduce help deliberately

Completion task: Audio transcript reads fifteen boxes; a clear written inventory says fifty boxes. The original audio is unavailable and the assistant reports fifteen as certain. Show the intermediate steps and give the conclusion with its scope.

Teacher check: Teacher explanation: There is an unresolved quantity conflict and no audio replay. Preserve both sources and ask the responsible person; certainty is unsupported.

Start by identifying the given inputs together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood. For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for Speech pipelines and multimodal evidence. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: A stage-level accuracy or authority policy does not establish every modality agrees or every downstream claim is supported.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step

for the class. If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial practice question

Which stage was initially wrong, and which stage was correctly operating on a wrong input?

Reasoned teacher answer

Speech recognition was wrong. The calculator correctly computed $30+7$; synthesis correctly repeated that wrong-for-intent result.

Changed-case reasoning: The calculator and spoken result should become 20 under the declared pipeline. $13+7=20$; the changed input removes the earlier recognition error.

Collect each learner's explanation before discussion. Use the first missing connection between input, deciding step and result to choose support. This question is worked-case practice, not fresh mastery evidence.

4. Run the weekly evidence lab

Run the weekly evidence lab

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.

Trace four audio failures stage by stage, locate the first mismatch, and propose a stage-specific test or fallback.

What did the hardest test teach you that the normal test did not?

Make the lab runnable

Use the concrete case as the shared normal case and the independent question as a second case. Each pair creates a boundary or missing-information case and predicts its result before testing. Keep the expected result separate from an observed mistake.

Record · What to capture

Record: Input or evidence

What to capture: Exact data, instruction or claim

Record: Expected result

What to capture: Answer or safe fallback declared before the run

Record: Observed result

What to capture: Actual result, including errors

Record: Explanation

What to capture: Rule, calculation or evidence explaining the outcome

Record: Next test

What to capture: One justified change and a new prediction

Diagnose and reteach

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation. Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

The worked result establishes every future case: A stage-level accuracy or authority policy does not establish every modality agrees or every downstream claim is supported.

A missing input may be replaced without recording the change: A replacement changes the evidence or contract. Record the change and its justification; do not silently turn unknown into a measured value.

Repeating an assessed case is new independent evidence: A replay checks consistency. An example used for feedback or revision cannot later be described as untouched final evaluation.

Match the support to the first error

What the response shows · Next teaching action

What the response shows: Missing or misread facts

Next teaching action: Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.

What the response shows: Wrong procedure

Next teaching action: Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.

What the response shows: Arithmetic or implementation error

Next teaching action: Trace intermediate values or lines; preserve the first mismatch and retry that step.

What the response shows: Correct result without a reason

Next teaching action: Ask for a trace and a changed case before marking independent understanding.

What the response shows: An unsupported wider claim

Next teaching action: Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal. Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

5. Assess and explain the answers

Assess and explain the answers

Worked case answer

Transcription supplied the wrong number. Confirm the transcript or ask for clarification. Correct arithmetic cannot repair wrong input.

Independent question

A recording says fifteen, but the transcript says fifty. The summary repeats fifty. Where did the error enter, and what should be checked first?

Expected independent answer

The transcription stage introduced the error. Compare the transcript with the recording before trusting or summarizing the number.

What counts as evidence

Criterion · 0 points · 1 point · 2 points

Criterion: Result

0 points: Incorrect or absent

1 point: Partly correct

2 points: Correct with necessary conditions

Criterion: Reasoning

0 points: No supporting trace

1 point: Some steps

2 points: Clear mechanism, calculation or source support

Criterion: Limit

0 points: Overclaims certainty

1 point: Vague limitation

2 points: Relevant uncertainty or missing fact

Criterion: Transfer

0 points: Copies without adapting

1 point: Adapts with prompts

2 points: Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold. Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the evidence that challenges this claim: “A summary can repair an upstream transcription error without checking the audio.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

The transcript changed sixteen to sixty. Display or read back the interpreted numbers before calculation when ambiguity matters. The intended result is 4; the calculator correctly computed its incorrect input. Retain only necessary diagnostic information.

Explanations for the additional workbook examples

W1 Explain the mechanism

Explain the weekly mechanism in your own words. Use the worked scenario to name its input, decision rule and output.

Teacher explanation: A multimodal pipeline may transcribe speech, inspect an image and generate a response. Errors at one stage can propagate even when later stages run correctly. Track source modality, segment/time references, transcription, interpretation and final claims separately. Compare conflicting evidence without assuming one modality always wins. Ask for clarification when the task needs unresolved information.

W2 Trace the worked case

Audio transcript reads fifteen boxes; a clear written inventory says fifty boxes. The original audio is unavailable and the assistant reports fifteen as certain. Show the intermediate steps and give the conclusion with its scope.

Teacher explanation: There is an unresolved quantity conflict and no audio replay. Preserve both sources and ask the responsible person; certainty is unsupported.

W3 Audit the limit

A report omits this qualification: A stage-level accuracy or authority policy does not establish every modality agrees or every downstream claim is supported. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Teacher explanation: A stage-level accuracy or authority policy does not establish every modality agrees or every downstream claim is supported. State the observed result separately from the unverified or out-of-scope claim; preserve the original evidence rather than inventing a missing fact.

W4 Apply a new case independently

Transcript says room E; a current written approval says room F. The task requires current approved room. Written approval is declared authoritative; trace the response and the transcription limit.

Teacher explanation: Report F from the authoritative approval, noting the transcript conflict if relevant. This resolves the task authority, not what the original speaker actually said.

W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison. Do not reuse W4 as a new final test after receiving feedback.

Teacher explanation: Accept a concrete variant of the worked input or a missing/conflicting/boundary condition, with an expectation derived from the declared procedure. Require preserved settings, original outputs and data roles. A stage-level accuracy or authority policy does not establish every modality agrees

or every downstream claim is supported. The new case should test this particular mechanism rather than merely restate the original result.

Assess independent transfer

Selected case: Known audio says “nineteen minus six”. Transcription says “ninety minus six”; the calculator returns 84. Locate the first error and propose a check.

Selected case answer

The first error is transcription: 19 became 90. The calculator correctly computes $90 - 6 = 84$ for its received input, but the intended result is 13. Display or read back the recognized expression before acting.

Dimension · Evidence required

Dimension: Explanation

Evidence required: Names the applicable idea or procedure and ties it to the supplied facts.

Dimension: Trace and result

Evidence required: Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.

Dimension: Limits

Evidence required: States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace. Do not penalise an honestly recorded model failure if the learner correctly explains its evidence and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case. Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

6. Connect and extend learning

Connect and extend learning

Verified Study Buddy

Show transcripts and require confirmation before saving memory or using tools.

Keep the decision, input, result, explanation and remaining limitation in the project evidence record.

Retrieve this record next week rather than restarting the project.

Support and stretch

Provide one short approved passage; separate instruction, input, output and evidence. Stretch: introduce an unsupported claim or conflicting source and validate claim by claim with a fallback.

Exit reflection

Ask for one decision the learner can now justify and one situation requiring more evidence or human review. Confidence ratings are separate from correct independent performance.

Bridge to the next week

Next: Week 13: Multimodal Evidence. Retrieve today's evidence and identify what carries forward and what is new.

Support and fresh reassessment

Provide an input/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace. Extension: compare two concrete variants of this mechanism while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

W4: Report F from the authoritative approval, noting the transcript conflict if relevant. This resolves the task authority, not what the original speaker actually said. Require intermediate reasoning and the stated scope. Fresh retry: Transcript says eight; a validated order record says eighteen and is declared authoritative for ordering. Expected reasoning: Use eighteen for the bounded order-record task; retain the transcript discrepancy and do not claim perfect transcription. If students need support, record it and reassess with another new case rather than awarding independence to a copied answer.

Next connection: Retrieval and indexing. Preserve the current case record and retrieve its evidence boundary before moving on.

Plan the delayed return

Suggested return points in this level: ai-3/13, ai-3/15, ai-3/18. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used. For a new level, revisit the previous capstone before checking the new prerequisites.

Record · Teacher evidence

Record: Learner code and date

Teacher evidence: _____

Record: Concept and case used; prior exposure

Teacher evidence: _____

Record: Explanation / trace / limits

Teacher evidence: _____

Record: Support and first repair

Teacher evidence: _____

Record: Changed case and delayed outcome

Teacher evidence: _____

Record: Next teaching action

Teacher evidence: _____

Use the evidence to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

Show Study Buddy recognised questions before calculating or retrieving important answers, especially numbers and dates.